

Sri Lanka Institute of Information Technology



SE2030 – Software Engineering
Year 2 Semester 1 – 2026

Web based – Shopping Mall
Proposal Document

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Group 07

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1. Introduction

The rapid growth of online retails has transformed the way businesses reach customers, moving away from physical stores and into digital marketplaces which are accessible anytime, anywhere. However, most of these digital marketplaces are built around a single seller whose managing their own catalogue, orders and a set of customers whom they interact with in isolation. This creates a distasteful experience for both the customers, and the merchants. Customers would need manage communication with each seller individually, visit multiple separate websites just to compare products and track each order separately. This would create an unpleasant situation for merchants, particularly small and medium sized sellers who often lacks an affordable, ready-made platform where they can list the product without having to build an entire system of their own.

To resolve this issue, a unified, multi-merchant platform must be addressed, by bringing several independent sellers together under one system, giving customers a single point of access to a wider range of products while giving the merchants a shared infrastructure to manage their business individually.

This project proposes the development of a web-based multi-merchant shopping mall system built using Java, Spring Boot and JSP with Microsoft SQL Server as the underlying database that allows multiple merchants to register and manage their own stores while customers to be given the ability to browse, purchase, and interact with products from different sellers, rate the products within a single cohesive platform. The system is designed to streamline the core processes of an online marketplace, from merchant and product management to cart and order handling, to reviews and customer support within one integrated application.

In addition, the project serves as a practical application of software engineering principles by following the Software Development Life Cycle (SDLC), adopting the Agile development methodology, and applying UML modelling techniques to design a well-structured, modular, and maintainable system.

1.2 Objectives

- To provide a centralized system for managing all **system actors and their roles** including registration, authentication, and role-based access for

customers, merchants, and admins, alongside merchant-specific capabilities such as store setup and product catalogue ownership.

- To provide customers with an organized, searchable product catalogue, allowing them to browse products across multiple merchants, filter by category and view detailed product information.
- To allow customers to collect and manage products of interest before purchasing through a shopping cart system supports adding, updating and removing items.
- To facilitate a secure and reliable order placement process that allows customers to complete purchase and receive confirmation with payment handled as part of the order flow.
- To deliver actionable feedback and reputation insights to both customers and merchants through verified product reviews, aggregated ratings, and merchant-facing analytics that support trust building and informed decision making.
- To establish a direct communication channel between customers and the platform/merchants through a support and inquiry system, so customers can raise issues and receive timely responses.

1.3 Stakeholders

There are 3 major stakeholders in this system. Merchants are business owners or sellers who use the platform to register their store list and manage their products and fulfil customer orders. Customers are the end users of the system who browse the product catalogue and place order. Finally, the Admin oversees the overall platform, ensuring merchants and content comply with platform policies and acts as an intermediary in resolving disputes or customer inquiries. Together, these three roles interact within the system to enable a functioning multi-merchant marketplace.

1.4 Scope and Limitations

In Scope

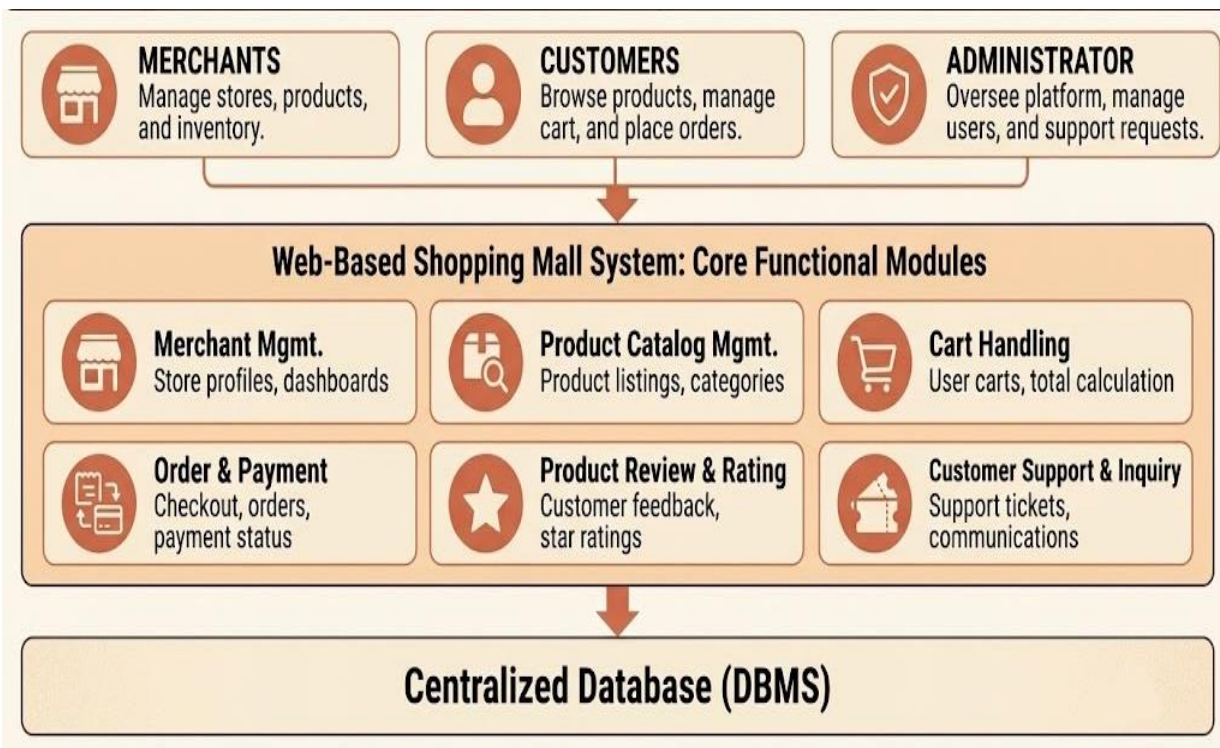
- Develop a responsive web application accessible through desktop and mobile browsers.
- The six core function, merchant management, product catalogue management, shopping cart handling, order and payment handling, product review and rating, and customer support/inquiry system
- Supporting minor functions needed to make the core functions work. (Eg: user registration and login (for both merchants and customers))

Out of Scope

- Real payment gateway integration
- Realtime delivery tracking
- Mobile application
- Merchant analytics

2. System overview diagram

The diagram shows the system's high-level architecture using a layered structure that separates user interaction, system processing, and data storage.



3. Functional Requirements

The system's functional requirements are organized according to the six main functions depicted above. These requirements describe the key features and capabilities that the system is planning to provide.

- The system shall allow merchants to register, verify, and manage their shop profiles.
- The system shall enforce role-based access, ensuring merchants, customers, and platform employees can only perform actions permitted for their role.
- The system shall allow regular customers to create personal accounts on the system.
- The system shall allow merchants to add, edit, and manage their product catalog.
- The system shall allow customers to add products to a cart and manage cart items before checkout.
- The system shall allow customers to complete secure payments and track orders through delivery.
- The system shall allow customers to submit ratings and reviews for products they have purchased, with reviews restricted to verified buyers.
- The system shall calculate and display aggregated rating data, including per-product distribution and overall merchant reputation scores.
- The system shall notify or flag products for platform employees when ratings fall below a defined threshold.
- The system shall allow customers to submit inquiries or complaints, and platform employees to track and resolve them.

4. Non-Functional Requirements

The following quality attributes describe the expected performance, security, and reliability standards of the system in addition to its functional requirements.

Quality Attribute	Description
Performance	The system should respond to user actions such as searching for the product catalogue or adding items to the cart within an acceptable time period, even as the number of products, merchants, and order

Quality Attribute	Description
	grows. Database queries against MSSQL should be optimized to keep catalogue browsing and search responsive.
Security	The system must protect user data and prevent common vulnerabilities. Passwords should be stored using hashing, user sessions should be managed securely, and inputs should be validated to prevent SQL injection. Especially given direct interaction with MSSQL. Merchants should only be able to manage their own products and orders, not those of other merchants.
Usability	The interface, built using JSP, should be simple and intuitive for both customers and merchants with minimal technical background. Navigation between browsing, cart, checkout, and support should be clear and consistent, minimizing the number of steps needed to complete common tasks like placing an order.
Reliability	The system should function consistently without unexpected crashes or data loss, particularly during critical operations like checkout and payment confirmations. Order and transactions must be always recorded accurately and retrievable.
Scalability	The system should be designed to support a growing number of merchants, products, and customers without requiring a major redesign.
Maintainability	The system should be structured using clear separation of concerns so that future developers can update or extend individual modules.

5. Major Stakeholders / Users.

Stakeholder	Role	Assigned Module(s)
Customer	Browse products, manage carts, place orders, make payments, track deliveries, and submit reviews/inquires.	Cart, Order & Payment, Reviews, Support
Merchant	Register and manage their shop profile, manages product catalogue, fulfills order	Merchant Management, Product Catalogue
Platform Employee	Approve/monitor merchants, oversee transactions, resolve customer support inquiries.	Merchant Management (oversight), Support

The system design is based on three main user roles, with each role supported by the six core functions. It provides customers with a simple self-service booking experience while helping administrators and staff manage accurate vehicle, branch, and maintenance records through proper system validations.

6. Six Major Functions.

3.1 User & Stakeholder Management

This function serves as the centralized identity and role administration engine for the platform, managing registration, verification, and access control for every type of system users - customers, merchants, and platform employees. Rather than treating account management as a background utility, this function governs how each role is onboarded, verified, and permitted to act within the system, forming the foundation that all other functions depend on.

- Allows customers to register and manage a personal account
- Allows merchants to register, submit shop details, and manage their shop profile

- Allows platform employees to verify, approve, or suspend merchant accounts to maintain marketplace trust
- Enforces role-based access control, ensuring each actor can only perform actions permitted for their role

3.2 Product catalogue management

This function provides customers with an organized and searchable view of product from all merchants on the platform, making it easy to discover items across categories and sellers.

- The system categorizes the products (e.g., electronics, clothing, groceries)
- Customers can search product by name and keyword
- Customers can filter product by category, price range, or merchant
- The system displays a detailed product page
- Customers can view products from multiple merchants under a unified catalogue view

3.3 Shopping cart handling

This function lets customers collect products they intend to purchase, adjust quantities and review their selections before proceeding to checkout.

- Customers can add products to the cart from the catalogue
- Customers can update the quantity of items in the cart
- Customers can remove the items from the cart
- The system displays the cart total before the checkout
- Customers can view the cart before checkout

3.4 Payment and Order tracking

This function manages the process of converting a customer's cart into a confirmed order, handling payment and allowing both customers and merchants to track order progress

- Customers can checkout process that converts cart contents into an order
- Payment option (redirects to the payment gateway)
- The system lets the customer know of the order confirmation and generates a summary

- Customers can track the order status (e.g., pending, processing, shipped, delivered)
- Customers can view their order history
- The system allows merchant-side order management (View orders placed for the product, update status)

3.5 Product Feedback & Reputation Analytics

This function gives customers a trustworthy way to evaluate products and merchants, while giving merchants and platform employees meaningful insight into product and shop performance. By combining verified feedback and proactive quality monitoring, this function turns customer feedback into a measurable reputation signal across the platform.

- Allows customers to submit a star rating and written review for a product, restricted to verified purchasers only
- Calculates and displays average rating and full rating distribution per product
- Aggregates individual product ratings into an overall merchant reputation score
- The system lists individual reviews on the product detail page
- The system restricts reviews only to the customers who purchased that product
- Allows merchants to publicly respond to customer reviews
- Automatically flags products for platform employee attention when ratings fall below a defined threshold

3.6 Customer support and inquiry system

This function gives customers a direct channel to raise inquiries or issues and receive timely responses, ensuring support is not limited to the point of purchase.

- Customers can submit inquiries or complaints tied to an order or product
- Administrators can respond to the submitted inquiry and connect with the merchant if it is necessary
- Customers can view the status of the submitted inquiry
- The system manages a basic FAQ section

7. Minor Functions

The following supporting functions help the six main functions operate smoothly. They are shared across multiple modules rather than belonging to a single function and will be developed as reusable components during implementation.

- User registration, secure login, password reset by email, and profile management
- Role-based access control (Customer, Merchant, Platform Employee) session timeout, and automatic logout for security
- Notifications for order status, delivery updates, low stock, and support ticket responses
- Search and filter across products, categories, shops, and orders
- Wishlist / save-for-later functionality for customers
- Admin dashboard aggregating platform-wide sales, active merchants, pending inquiries, and system alerts

8. System Limitations

8.1 Platform / Technical Constraints

- The system is a web-based application, which is only accessible through standard desktop/laptop browsers. No native mobile application is provided. While the interface may be responsive to some extent, it is not optimized for small mobile screens.
- The system is built using Java, Spring Boot, and JSP, requiring a compatible Java runtime environment and application server for deployment.
- The system uses Microsoft SQL Server as its database, meaning deployment requires an environment with MSSQL support therefore is not designed to run on other database engines without modification.
- The system is developed and tested in a local/development environment for academic purposes and is not deployed to a production grade or cloud hosted environment with real world traffic handling.

8.2 Functional Limitations

- Payment processing is simulated, not integrated with a real third-party payment gateway therefore no actual monetary transactions occur.
- Delivery is not tracked in real time - order status updates are manually set by merchants rather than integrated with courier/shipping APIs.
- The system supports a single language and currency. Multi-language and multi-currency support are not included.
- Product feedback analytics are rule-based, not powered by machine learning. Low-rating flagging uses a fixed threshold rather than predictive or sentiment-based analysis.
- The system does not include advanced features such as personalized product recommendations, dynamic pricing, or AI-driven search.

8.3 Assumptions

- Merchants are assumed to provide accurate and truthful information during registration. The system's verification step is limited to basic checks and does not perform legal/business registration validation.
- Customers are assumed to have basic digital literacy and access to a device with a modern web browser and internet connection.
- The system assumes a moderate number of concurrent users appropriate for an academic project demo, not enterprise-scale traffic.
- Product and order data used during development/testing may be sample/dummy data rather than real merchant inventories.

9. Project Timeline

9.1 Development Methodology

The team will adopt an Agile development approach, building the system through short, iterative development cycles. A product backlog will be created based on the six major functions identified in Section 7, with tasks prioritized according to their complexity and dependency on other features. Each sprint will focus on developing and refining a

selected set of functions, followed by a sprint review to assess progress, identify issues, and plan subsequent iterations. Progress updates, completed work, and plans for upcoming sprints will be documented in the Agile Sprint Summary, to be included in the Week 10 design document.

9.2 Weekly Plan

The table below shows the planned activities and timeline of the group from the proposal submission stage until the final presentation.

Week	Planned Activities
3	Planning
4 – 5	System design
6 – 8	Development
9	Integration
10	Testing
11 – 12	Deployment
13	Final presentation and viva.
14	Report documentation, final submission.

10. Conclusion

The web-based shopping mall brings together independent merchants and customers on a single, accessible platform, replacing fragmented, manual processes with a structured system for browsing, purchasing, and after sales support. By focusing on the six core functions, user and stakeholder management, product catalogue management, shopping cart handling, order and payment handling, product feedback and reputation analytics,

and customer support, the system addresses the essential needs of its three stakeholder groups: customers, merchants, and platform employees. The group has divided responsibility for each core function among its six members, with a clear week-by-week plan from Week 3 through Week 14 covering design, development, integration, and testing. The team is confident that, by following this plan, the project can be completed successfully and deliver a functional, reliable, and user-friendly online shopping mall.